

## Introduction

Random-slope models allow each cluster its own slope for one or more predictors, capturing between-cluster heterogeneity in how a covariate affects the outcome. Where random-intercept models assume that all clusters share the same effect of  $X$  but differ in baseline, random-slope models drop the shared-slope assumption and let the slope vary. They are essential for any longitudinal analysis where individual trajectories diverge over time and where the population-average slope may hide substantial subject-to-subject variability.

## Prerequisites

A working understanding of random-intercept models, mixed-effects model syntax in `lme4`, and the bivariate Normal distribution for joint random effects.

## Theory

For a predictor  $X$  varying within clusters:

$$Y_{ij} = (\beta_0 + u_{0i}) + (\beta_1 + u_{1i})X_{ij} + \varepsilon_{ij},$$

with  $(u_{0i}, u_{1i})^\top \sim \mathcal{N}(0, \Sigma)$  where  $\Sigma$  contains intercept variance  $\sigma_0^2$ , slope variance  $\sigma_1^2$ , and correlation  $\rho_{01}$ . The “correlated random slope” formulation (`X | cluster`) estimates all three parameters jointly; the “uncorrelated” form (`X || cluster`) constrains  $\rho_{01} = 0$ .

A non-zero  $\rho_{01}$  indicates systematic dependence between baseline level and rate of change — for instance, subjects who start higher progress faster (positive correlation) or slower (negative correlation).

## Assumptions

Random effects bivariate Normal, independent across clusters; conditional Normal residuals; the slope variation is real and not driven by a few influential clusters.

## R Implementation

```
library(lme4)

data(sleepstudy, package = "lme4")

# Random intercept + slope (correlated)
fit_rs <- lmer(Reaction ~ Days + (Days | Subject), data = sleepstudy)
summary(fit_rs)

# Uncorrelated random slope
fit_rs_unc <- lmer(Reaction ~ Days + (1 | Subject) + (0 + Days | Subject),
                  data = sleepstudy)

# Compare with random-intercept-only
fit_ri <- lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy)
anova(fit_ri, fit_rs)
```

## Output & Results

`summary(fit_rs)` reports the population-level fixed effects, the standard deviations of the random intercept and random slope, and their correlation. `anova()` between nested models compares them via a likelihood-ratio test (with the boundary-mixture reference for variance components).

## Interpretation

A reporting sentence: “Adding a random slope on `Days` substantially improved fit ( $\chi^2(2) = 42.1$ ,  $p < 0.001$ ); between-subject standard deviation of the slope was 5.9 ms per day, with intercept-slope correlation 0.07. The population-level slope was 10.5 ms per day (95 % CI 7.6 to 13.4).” Always report the random-slope variance and the intercept-slope correlation alongside fixed effects.

## Practical Tips

- Random slopes require enough observations per cluster (rule of thumb: at least 10 per cluster); with fewer, the slope variance is unstable and may produce singular fits.
- Near-zero random-slope variance signals over-specification; drop the random slope or constrain  $\rho_{01} = 0$  via the `||` syntax.
- Centring the predictor (`X - mean(X)`) often reduces intercept-slope correlation that arises mechanically from the parameterisation rather than the data.
- Check the random-effects variance-covariance with `VarCorr(fit)`; correlation estimates near  $\pm 1$  are a red flag for over-fitting.
- Use parametric bootstrap (`pbkrtest::PBmodcomp`) for small-sample LRTs on random-slope variance; the asymptotic  $\chi^2$  mixture can be unreliable.
- For non-Gaussian outcomes, the same syntax extends via `glmer()` with the appropriate family.

## R Packages Used

`lme4` for `lmer()` and `glmer()`; `lmerTest` for  $p$ -values on fixed effects; `pbkrtest` for parametric-bootstrap LRTs on variance components; `glmmTMB` for an alternative implementation that handles complex error structures; `merTools` for prediction intervals that respect random-slope uncertainty.

## For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

## See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (`emmeans`)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures  $\rightarrow$  mixed models
- GAMs with `mgcv`
- Non-linear regression with `nls`
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean

- Decision curves, NRI, IDI